

STAC

SpatioTemporal Asset Catalogs

Phi Week 2019 - Frascati

Ariel Walcutt: @GISandToast

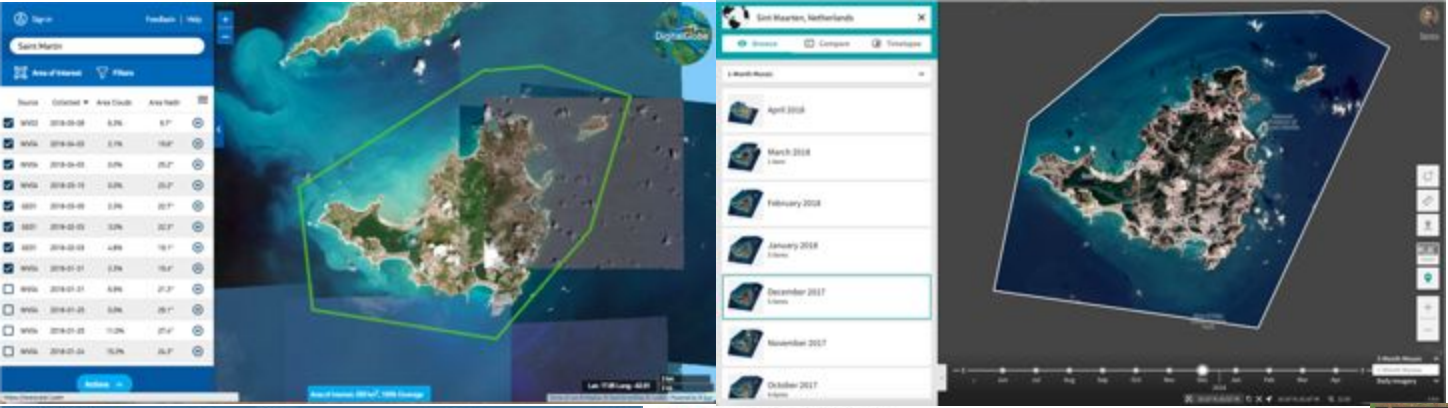
Matthew Hanson: @GeoSkeptic

Element 84

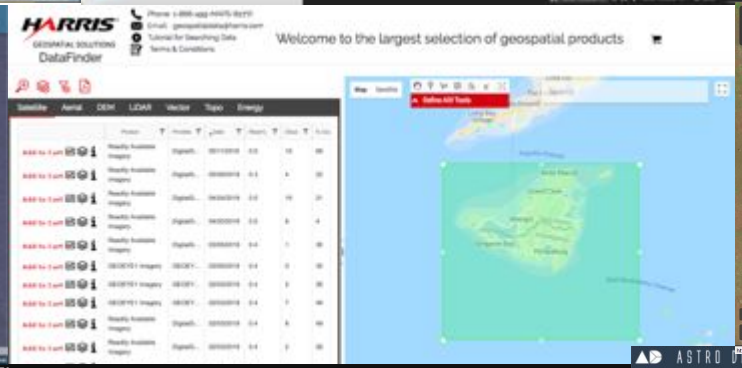
@Element84

Discoverability of large geospatial datasets is difficult

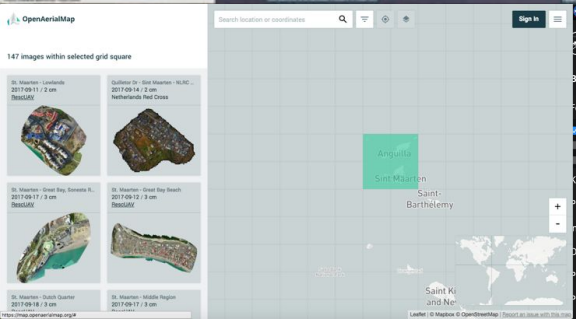
Every data provider has their own data portal


Saint Martin
Sint Maarten, Netherlands
April 2018
March 2018
February 2018
January 2018
December 2017
November 2017
October 2017

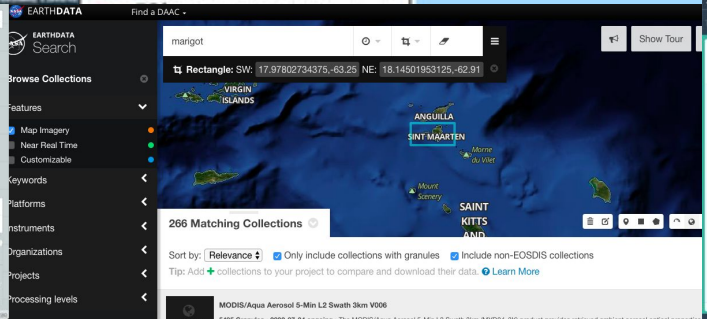

SENTINEL Hub Playground

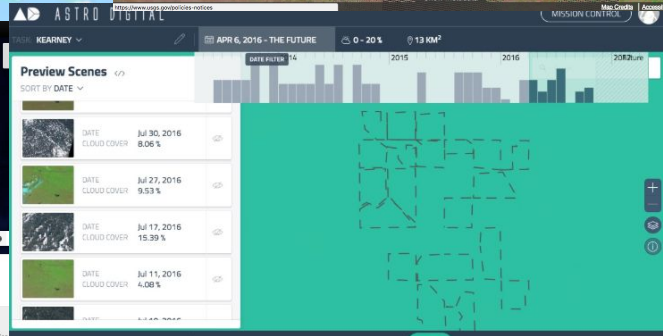

HARRIS GEOSPATIAL SOLUTIONS
DataFinder
Welcome to the largest selection of geospatial products


USGS LandsatLook
Show Metadata


OpenAerialMap
147 images within selected grid square

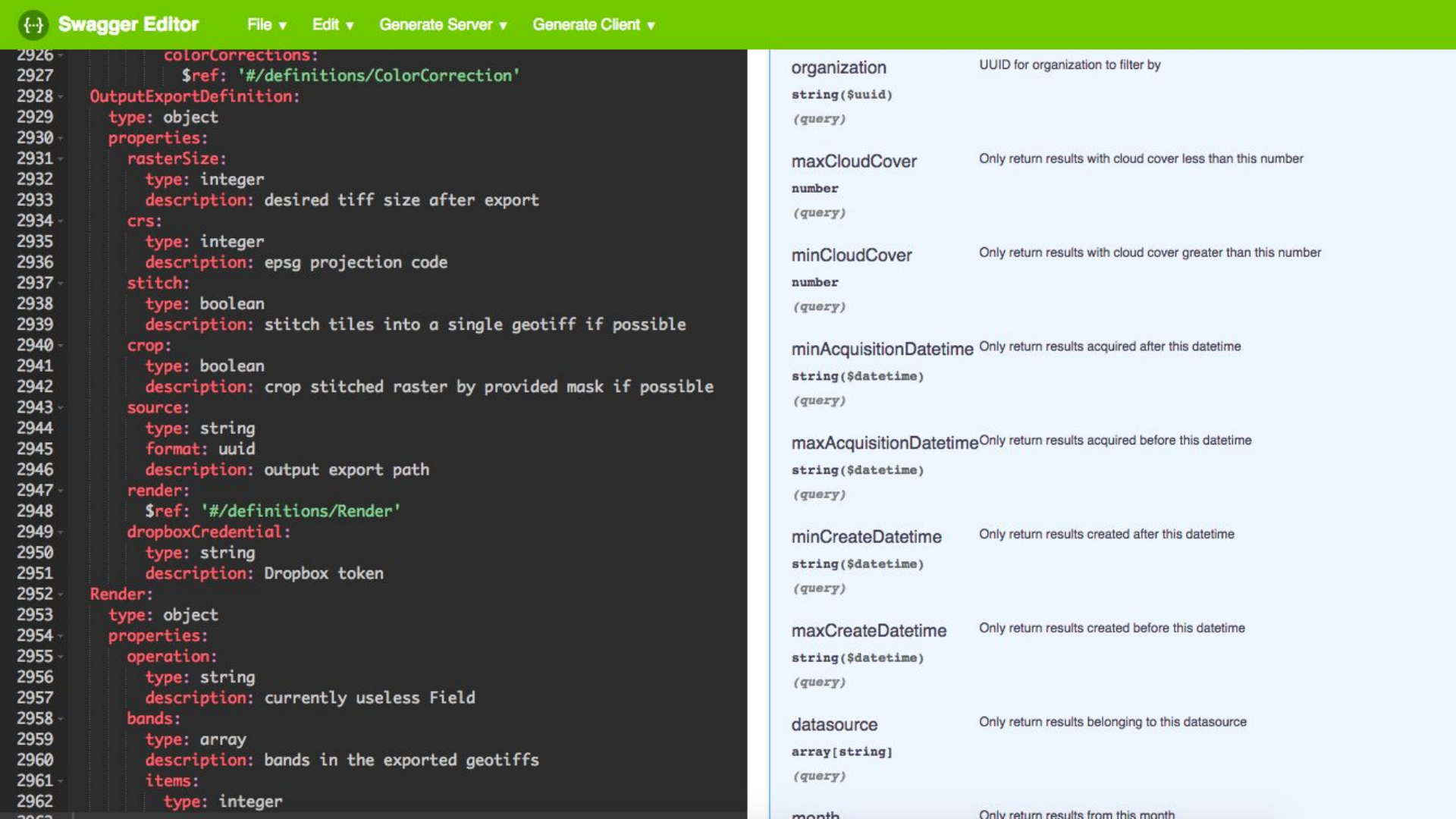
Anguilla
Sint Maarten
Saint-Barthelemy


EARTHDATA
Search
marigot
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
Show Tour
Browse Collections
Features
Map Imagery
Near Real Time
Customizable
Keywords
Platforms
Instruments
Organizations
Projects
Processing levels
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006


ASTRO DIGITAL
MISSION CONTROL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
Jul 30, 2016
8.06 %
Jul 27, 2016
9.93 %
Jul 17, 2016
15.39 %
Jul 11, 2016
4.08 %

APIs are even more different

Different field names, different queries


Swagger Editor
File
Edit
Generate Server
Generate Client
colorCorrections:
$ref: '#/definitions/ColorCorrection'
OutputExportDefinition:
type: object
properties:
rasterSize:
type: integer
description: desired tiff size after export
crs:
type: integer
description: epsg projection code
stitch:
type: boolean
description: stitch tiles into a single geotiff if possible
crop:
type: boolean
description: crop stitched raster by provided mask if possible
source:
type: string
format: uuid
description: output export path
render:
$ref: '#/definitions/Render'
dropboxCredential:
type: string
description: Dropbox token
Render:
type: object
properties:
operation:
type: string
description: currently useless Field
bands:
type: array
description: bands in the exported geotiffs
items:
type: integer
organization
UUID for organization to filter by
string($uuid)
(query)
maxCloudCover
Only return results with cloud cover less than this number
number
(query)
minCloudCover
Only return results with cloud cover greater than this number
number
(query)
minAcquisitionDatetime
Only return results acquired after this datetime
string($datetime)
(query)
maxAcquisitionDatetime
Only return results acquired before this datetime
string($datetime)
(query)
minCreateDatetime
Only return results created after this datetime
string($datetime)
(query)
maxCreateDatetime
Only return results created before this datetime
string($datetime)
(query)
datasource
Only return results belonging to this datasource
array[string]
(query)

What is STAC?

SpatioTemporal Asset Catalog

- JSON, GeoJSON

What is STAC?

SpatioTemporal Asset Catalog

- Single metadata model for geospatial data
  - Focus on search and discovery
  - Simple and extensible
- Linked static catalogs spec
  - Crawlable
- Dynamic API spec (OGC API)
  - Searchable on arbitrary fields

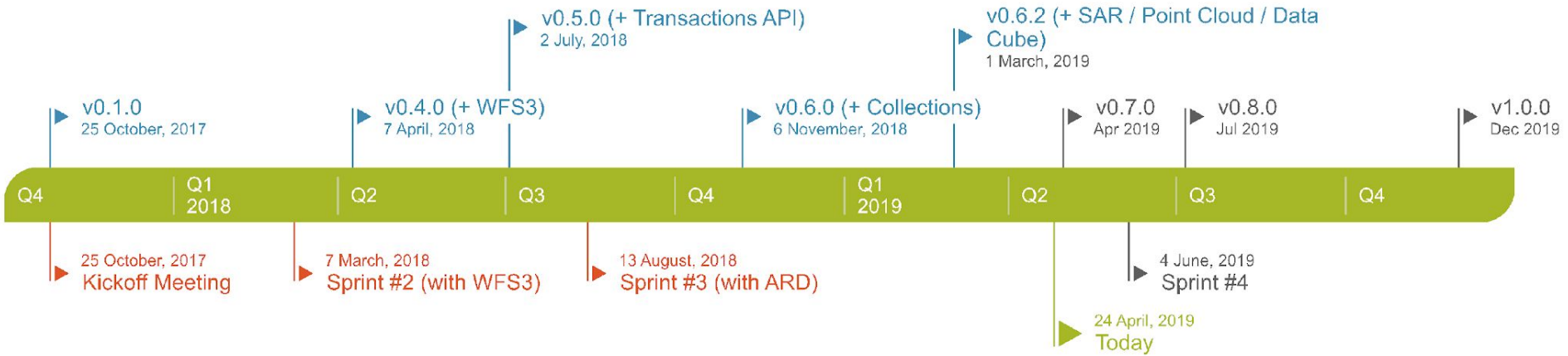
What STAC is not

SpatioTemporal Asset Catalog

- Full-fledged metadata standard
- Replacement for data provider's internal metadata
- Single source of truth
- Covering *all* kind of datasets
- Replacement for ISO 19115, OGC CSW, …

SpatioTemporal Asset Catalog

When?



stacspec.org



SpatioTemporal Asset Catalog

Who?

- 29 contributors on github
- 10 with 10+ commits



Radiant Earth Foundation

EARTH IMAGERY FOR IMPACT

Element 84

DigitalGlobe

azavea



developmentSEED

planet.



openEO

HARRIS



Australian Government
Geoscience Australia



Google Earth Engine



HEXAGON GEOSPATIAL



Boundless



SpaceNet™

STAC Entities

- Catalogs
  - There's always a root catalog
  - Contains other catalogs and collections
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Ultimately contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when and where)

Catalog Fields

Element	Type	Description
stac_version	string	**REQUIRED.** The STAC version the catalog implements.
id	string	**REQUIRED.** Identifier for the catalog.
title	string	A short descriptive one-line title for the catalog.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the catalog. CommonMark 0.28 syntax MAY be used for rich text representation.
links	[Link Object]	**REQUIRED.** A list of references to other documents.

Links

Field Name	Type	Description
href	string	**REQUIRED.** The actual link in the format of an URL. Relative and absolute links are both allowed.
rel	string	**REQUIRED.** Relationship between the current document and the linked document. See chapter "Relation types" for more information.
type	string	Media type of the referenced entity.
title	string	A human readable title to be used in rendered displays of the link.

Hierarchical Links

Type	Description
self	STRONGLY RECOMMENDED. *Absolute* URL to the location that the catalog file can be found online, if available. This is particularly useful when in a download package that includes metadata, so that the downstream user can know where the data has come from.
root	STRONGLY RECOMMENDED. URL to the root STAC Catalog or Collection. Catalogs should include a link to their root, even if it's the root and points to itself.
parent	URL to the parent STAC Catalog or Collection. Non-root catalogs should include a link to their parent.
child	URL to a child STAC Catalog or Collection.
item	URL to a STAC Item.

Collection Fields

Element	Type	Description
stac_version	string	**REQUIRED.** The STAC version the collection implements.
id	string	**REQUIRED.** Identifier for the collection that is unique across the provider.
title	string	A short descriptive one-line title for the collection.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the collection. CommonMark 0.28 syntax MAY be used for rich text representation.
keywords	[string]	List of keywords describing the collection.
version	string	Version of the collection.
license	string	**REQUIRED.** Collection's license(s) as a SPDX License identifier or expression or `proprietary` if the license is not on the SPDX license list. Proprietary licensed data SHOULD add a link to the license text, see the `license` relation type.
providers	[Provider Object]	A list of providers, which may include all organizations capturing or processing the data or the hosting provider. Providers should be listed in chronological order with the most recent provider being the last element of the list.
extent	Extent Object	**REQUIRED.** Spatial and temporal extents.
properties	object	Common fields across referenced items. May also be used to describe standalone collections better that don't reference any items. See the section 'Common Fields' for more information.
links	[Link Object]	**REQUIRED.** A list of references to other documents.

Item Fields

Field Name	Type	Description
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. As most geospatial assets are already defined by some identification scheme by the data provider it is recommended to simply use that ID. Data providers are advised to include sufficient information to make their IDs globally unique, including things like unique satellite IDs.
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
geometry	GeoJSON Geometry Object	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED.** Bounding Box of the asset represented by this item using either 2D or 3D geometries. The length of the array must be 2*n where n is the number of dimensions. The array contains all axes of the southwesterly most extent followed by all axes of the northeasterly most extent specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84. When using 3D geometries, the elevation of the southwesterly most extent is the minimum depth/height in meters and the elevation of the northeasterly most extent is the maximum. This field enables more naive clients to easily index and search geospatially. STAC compliant APIs are required to compute intersection operations with the item's geometry field, not its bbox.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is required.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key. Some pre-defined keys are listed in the chapter 'Asset types'.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present. This field provides an easy way for a user to search for any Items that belong in a specified Collection.

Item Properties

Field Name	Type	Description
datetime	string	**REQUIRED.** The searchable date and time of the assets, in UTC. It is formatted according to RFC 3339, section 5.6.
license	string	Item's license(s) as a SPDX License identifier or expression or `proprietary` if the license is not on the SPDX license list. Proprietary licensed data SHOULD add a link to the license text, see the `license` relation type.
providers	[Provider Object]	A list of providers, which may include all organizations capturing or processing the data or the hosting provider. Providers should be listed in chronological order with the most recent provider being the last element of the list.
title	string	A human readable title describing the item.
created	string	Creation date and time of this metadata file. This is NOT the timestamp the asset was created. MUST be formatted according to RFC 3339, section 5.6.
updated	string	Date and time this metadata file was updated last. This is NOT the timestamp the asset was updated last. MUST be formatted according to RFC 3339, section 5.6.

Item Assets

Field Name	Type	Description
href	string	**REQUIRED.** Link to the asset object. Relative and absolute links are both allowed.
title	string	The displayed title for clients and users.
type	string	Media type of the asset (see chapter on Media Types below).

Not Secure | landsat-stac.s3.amazonaws.com/catalog.json

Apps | Devseed | Beachfront | Gaming | Imported | sort | Geospatial | Remote Sensing | Screens | Programming

+ - View source


```
{
   id: "landsat-stac",
   stac_version: "0.6.0",
   description: "STAC for Landsat data",
 - links: [
    - {
         rel: "self",
         href: "https://landsat-stac.s3.amazonaws.com/catalog.json"
      },
    - {
         rel: "root",
         href: "./catalog.json"
      },
    - {
         rel: "child",
         href: "landsat-8-l1/catalog.json"
      }
   ]
}
```


inks

Planet Disaster Data (planet-disaster-data)

Version 1.0.0

https://storage.googleapis.com/pdd-stac/disasters/catalog.json

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian organizations, and other coordinating bodies in support of the International Charter for Space and Major Disasters. Data is released for individual disaster events, providing a 30 day window pre- and post-disaster. Imagery is provided under Creative Commons licenses, free of charge, with either CC-BY-SA or CC-BY-NC.

Catalogs

Title
Hurricane Harvey



Provider

Planet Disaster Team <disaster-team@planet.com> *(producer, licensor)*

STAC Version	0.6.0
Keywords	disaster, open
License	CC-BY-SA-4.0
Temporal Extent	8/28/2017, 11:00:00 AM - now

Powered by STAC Browser

https://earth-search.aws.element84.com/stac

+ - View source


```
{
   stac_version: "0.6.2",
   id: "earth-search",
   title: "Earth on AWS STAC API",
   description: "A STAC API powered by sat-api for Earth on AWS datasets by Element 84",
 - links: [
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-awfi"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan10m"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-mux"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/sentinel-2-l1c"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/landsat-8-l1"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan5m"
       },
     - {
           rel: "self",
           href: "https://earth-search.aws.element84.com/stac"
       }
   ]
}
```


links

Content Extensions

Extension Name (Prefix)	Scope	Description	Maturity
Asset Definition (-)	Collection	Provides a way to specify details about what assets may be found in Items belonging to a collection	*Proposal*
Checksum (`checksum`)	Item +Catalog +Collection	Provides a way to specify file checksums for assets and links in Items, Catalogs and Collections.	*Proposal*
Data Cube (`cube`)	Item	Data Cube related metadata, especially to describe their dimensions.	*Proposal*
Datetime Range (`dtr`)	Item	An extension to provide datetime ranges with a start and an end datetime stamp in a consistent way.	*Proposal*
EO (`eo`)	Item	Covers electro-optical data that represents a snapshot of the earth for a single date and time. It could consist of multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands. The extension provides common fields like bands, cloud cover, off nadir, sun angle + elevation, gsd and more.	*Pilot*
Label (`label`)	Item	Items that relate labeled AOIs with source imagery	*Proposal*
Point Cloud (`pc`)	Item	Provides a way to describe point cloud datasets. The point clouds can come from either active or passive sensors, and data is frequently acquired using tools such as LiDAR or coincidence-matched imagery.	*Proposal*
SAR (`sar`)	Item	Covers synthetic-aperture radar data that represents a snapshot of the earth for a single date and time.	*Proposal*
Scientific (`sci`)	Item +Collection	Scientific metadata is considered to be data that indicate from which publication data originates and how the data itself should be cited or referenced.	*Proposal*
Single File STAC (-)	ItemCollection	An extension to provide a set of Collections and Items as a single file catalog.	*Proposal*

EO Extension

Field Name	Type	Description
eo:gsd	number	**REQUIRED.** Ground Sample Distance at the sensor.
eo:platform	string	**REQUIRED.** Unique name of the specific platform the instrument is attached to. For satellites this would be the name of the satellite (e.g., landsat-8, sentinel-2A), whereas for drones this would be a unique name for the drone.
eo:constellation	string	Name of the constellation that the platform belongs to. See below for details.
eo:instrument	string	**REQUIRED.** Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
eo:bands	[Band Object]	**REQUIRED.** This is a list of the available bands where each item is a Band Object.
eo:epsg	integer|null	EPSG code of the datasource, `null` if no EPSG code.
eo:cloud_cover	number	Estimate of cloud cover as a percentage (0-100) of the entire scene. If not available the field should not be provided.
eo:off_nadir	number	Viewing angle. The angle from the sensor between nadir (straight down) and the scene center. Measured in degrees (0-90).
eo:azimuth	number	Viewing azimuth angle. The angle measured from the sub-satellite point (point on the ground below the platform) between the scene center and true north. Measured clockwise from north in degrees (0-360).
eo:sun_azimuth	number	Sun azimuth angle. From the scene center point on the ground, this is the angle between truth north and the sun. Measured clockwise in degrees (0-360).
eo:sun_elevation	number	Sun elevation angle. The angle from the tangent of the scene center point to the sun. Measured from the horizon in degrees (0-90).

EO Band Object

Field Name	Type	Description
name	string	The name of the band (e.g., "B01", "B02", "B1", "B5", "QA").
common_name	string	The name commonly used to refer to the band to make it easier to search for bands across instruments. See below for a list of accepted common names.
description	string	Description to fully explain the band. CommonMark 0.28 syntax MAY be used for rich text representation.
gsd	number	Ground Sample Distance, the nominal distance between pixel centers available, in meters. Defaults to `eo:gsd` if not provided.
accuracy	number	The expected error between the measured location and the true location of a pixel, in meters on the ground.
center_wavelength	number	The center wavelength of the band, in micrometers (μm).
full_width_half_max	number	Full width at half maximum (FWHM). The width of the band, as measured at half the maximum transmission, in micrometers (μm).

EO Band Common Names

Common Name	Band Range (µm)	Landsat 5/7	Landsat 8	Sentinel 2	MODIS
coastal	0.40 - 0.45		1	1	
blue	0.45 - 0.50	1	2	2	3
green	0.50 - 0.60	2	3	3	4
red	0.60 - 0.70	3	4	4	1
yellow	0.58 - 0.62				
pan	0.50 - 0.70	8 (*L7 only*)	8		
rededge	0.70 - 0.75				
nir	0.75 - 1.00	4		8	2
nir08	0.75 - 0.90		5	8a	
nir09	0.85 - 1.05			9	
cirrus	1.35 - 1.40		9	10	26
swir16	1.55 - 1.75	5	6	11	6
swir22	2.10 - 2.30	7	7	12	7
lwir	10.5 - 12.5	6			
lwir11	10.5 - 11.5		10		31
lwir12	11.5 - 12.5		11		32

Linking Assets to Bands

```
- eo:bands: [
    - {
        name: "B1",
        common_name: "coastal",
        gsd: 30,
        center_wavelength: 0.44,
        full_width_half_max: 0.02
      },
    - {
        name: "B2",
        common_name: "blue",
        gsd: 30,
        center_wavelength: 0.48,
        full_width_half_max: 0.06
      },
    - {
        name: "B3",
        common_name: "green",
        gsd: 30,
        center_wavelength: 0.56,
        full_width_half_max: 0.06
      },
```

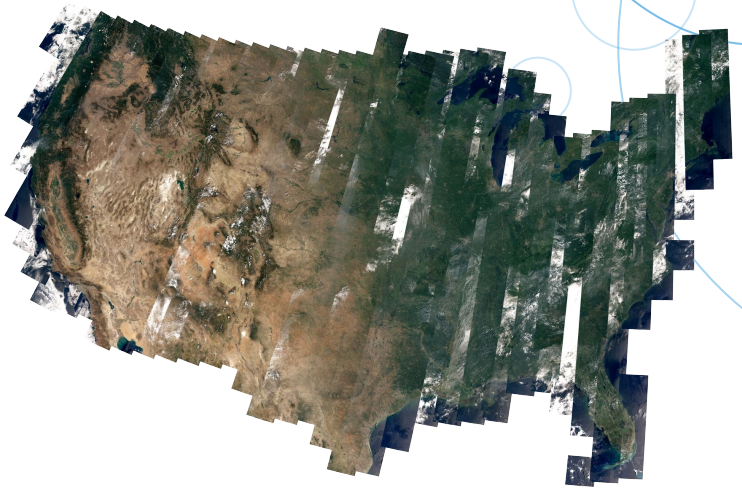
```
- assets: {
    - index: {
        type: "text/html",
        title: "HTML index page"
      },
    - thumbnail: {
        title: "Thumbnail image",
        type: "image/jpeg"
      },
    - B1: {
        type: "image/x.geotiff",
      - eo:bands: [
            0
        ],
        title: "Band 1 (coastal)"
      },
    - B2: {
        type: "image/x.geotiff",
      - eo:bands: [
            1
        ],
        title: "Band 2 (blue)"
      },
    - B3: {
        type: "image/x.geotiff",
      - eo:bands: [
            2
        ],
        title: "Band 3 (green)"
      },
```

E84 Earth Search

https://earth-search.aws.element84.com/stac

Public Search and Discovery API for Earth on AWS Datasets

- Central search catalog and standardized metadata
- Single API that allows searching of STAC metadata for all Earth on AWS datasets
- Powered by sat-api (https://github.com/sat-utils/sat-api)
- Backed by an ElasticSearch index
- Proposed initial datasets
  - Sentinel-1
  - Sentinel-2
  - CBERS-4
  - Landsat-8
  - NAIP



Landsat-8

Landsat is perhaps one of the most oft-cited and used EO datasets because of its historical archive (Landsat-8 was launched in 2013, but the entire Landsat archive goes back to 1982, with some caveats).

sat-search

https://github.com/sat-utils/sat-search

- Python library for searching STAC complaint endpoints
- Command Line Interface (CLI) as a general purpose tool
- Save search results to be loaded and used later
- Download specific assets from all items
- Specify specific band colors to download those bands
    - e.g., Download all “red” and “nir” bands from these Landsat and Sentinel Items

```
$ sat-search search --intersects maine.geojson --found
1674 items found
$ clear
```

SAT-API Browser

BUILD QUERY | RESULTS | CANDIDATES

Date Range

Start Date (required): 2019/07/14 06:33 PM

End Date (required): 2019/07/15 06:33 PM

Area Of Interest (required)

DRAW BBOX ON MAP

Property Filters

Add new filter

Select a property to filter

SUBMIT QUERY



NASA CMR STAC

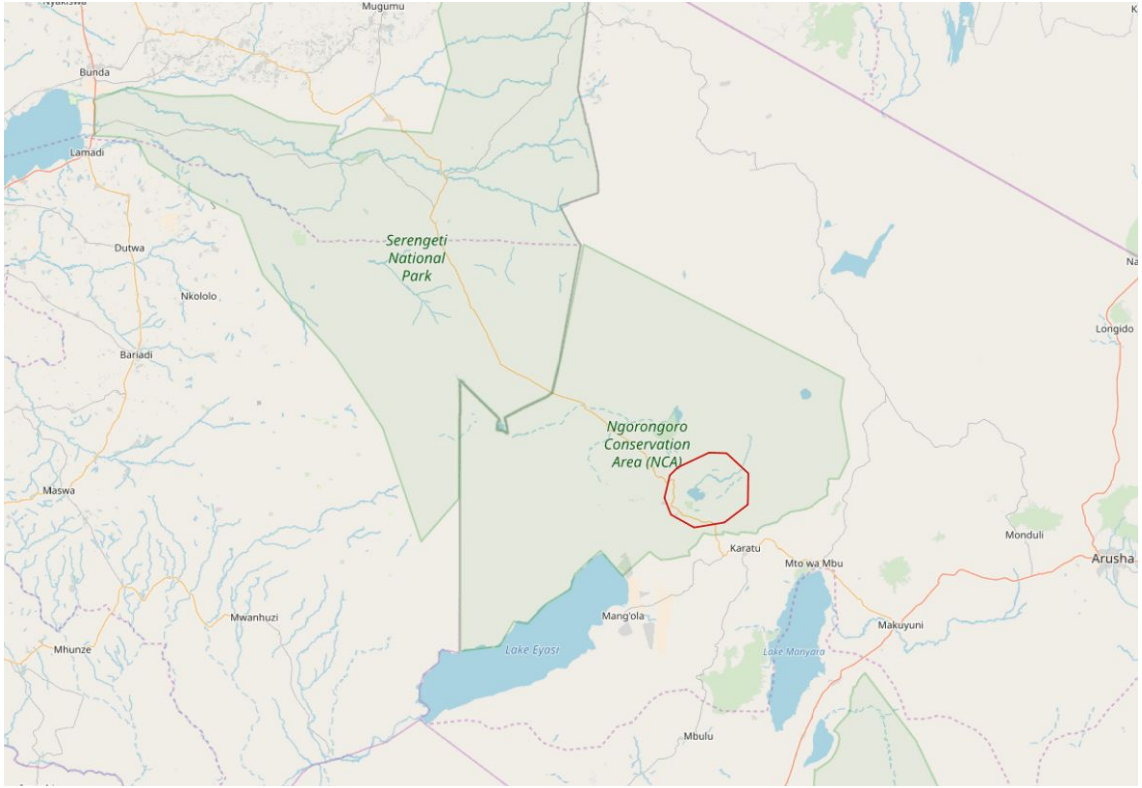
- NASA Common Metadata Repository
  - All NASA data from all DAACs in CMR … (?)
- Element 84 currently working on adding STAC support to CMR
  - By October
- Implemented as a proxy to CMR
  - Separate STAC API endpoint
  - no ingestion
  - Targeting STAC 0.8.0 (to be released by end of month)

sat-fetch

https://github.com/sat-utils/sat-fetch

- Works just like sat-search, plus...
- Option to fetch imagery for just AOIs, rather than the entire asset
- Requires GDAL, or use publicly available Docker image
- *Warning: sat-fetch is currently out of date*

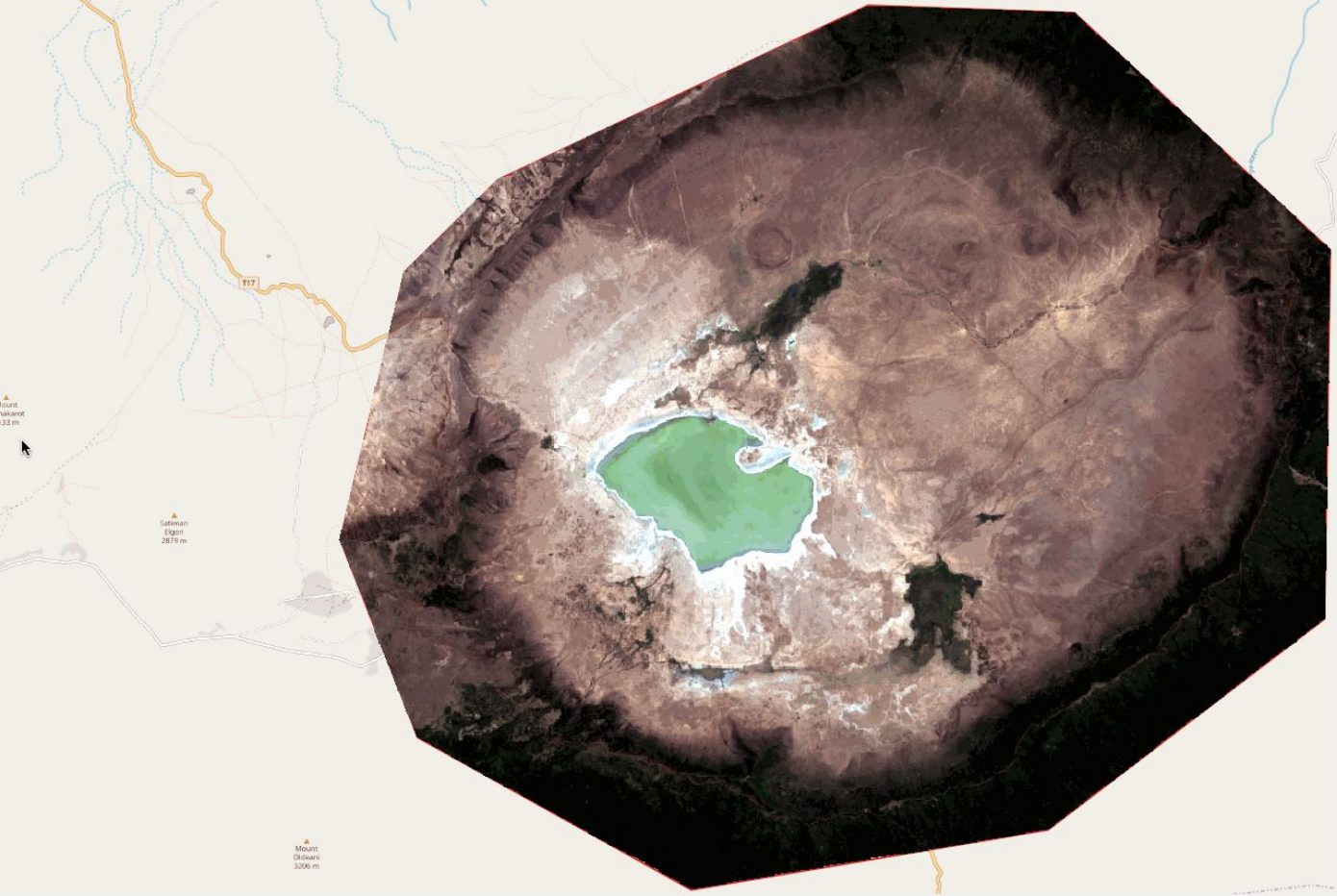
Ngorongoro crater: landsat-8 and sentinel-2



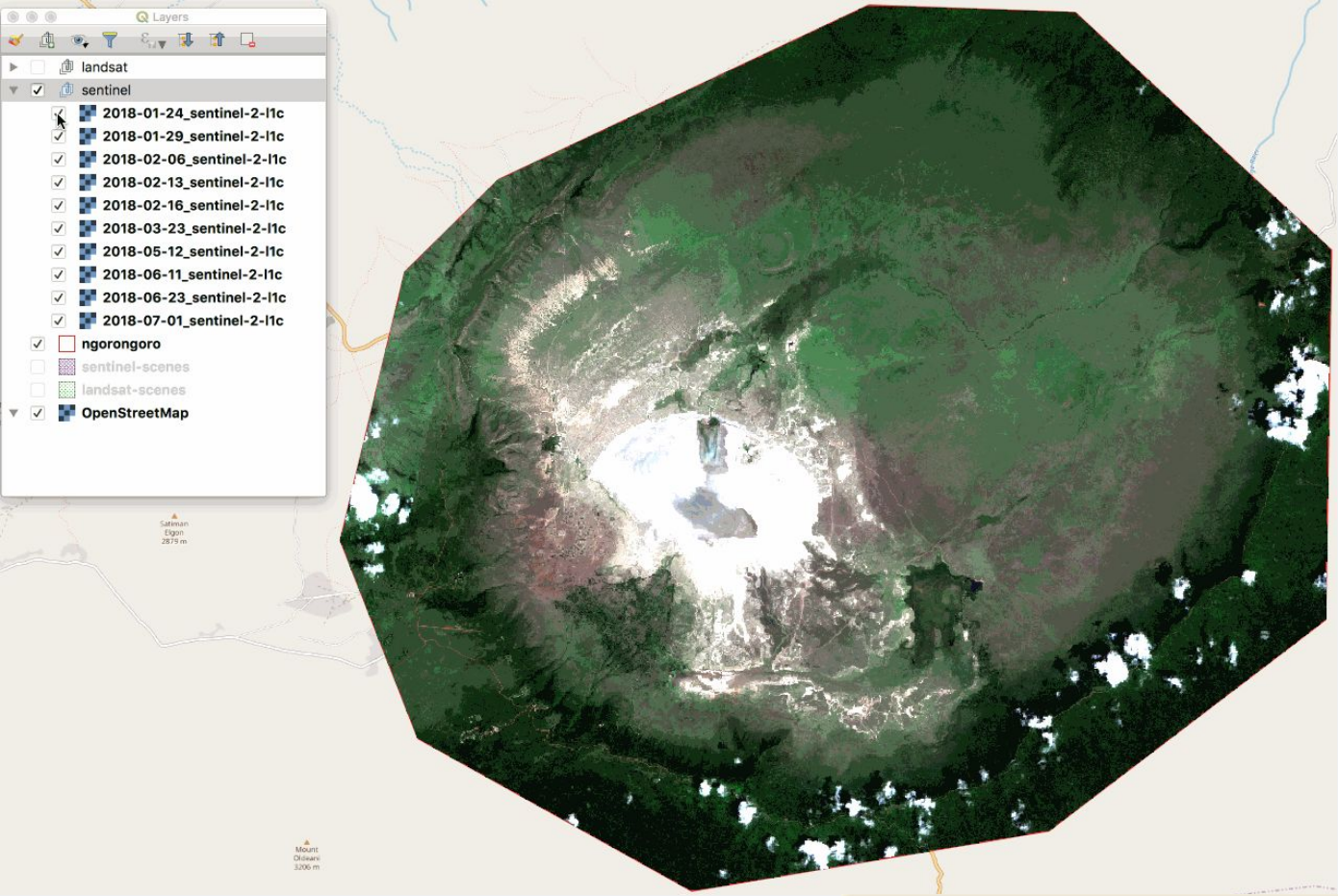
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-12-11  5  LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1  LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0  LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2  LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1  LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0  LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0  LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2  LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0  LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3  LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0  LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2  LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0  LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3  LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```




```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```




```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
real 8m6.525s
```